National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

Nasir Mahmood

Submitted by:

Afia Maham

Reg number:

23-NTU-CS-1126

Lab no:

1

Semester:

5th

Operating Systems – COC 3071L

SE 5th A – Fall 2025

Objective

The purpose of this assignment is to:

1. Configure **Ubuntu** inside **WSL2 (Windows Subsystem for Linux v2)**.
2. Install and configure **Git** in Ubuntu.
3. Generate and set up **SSH keys** to connect with GitHub.
4. Install the **C development environment** in Ubuntu.
5. Write a **Hello World** program in `C`.

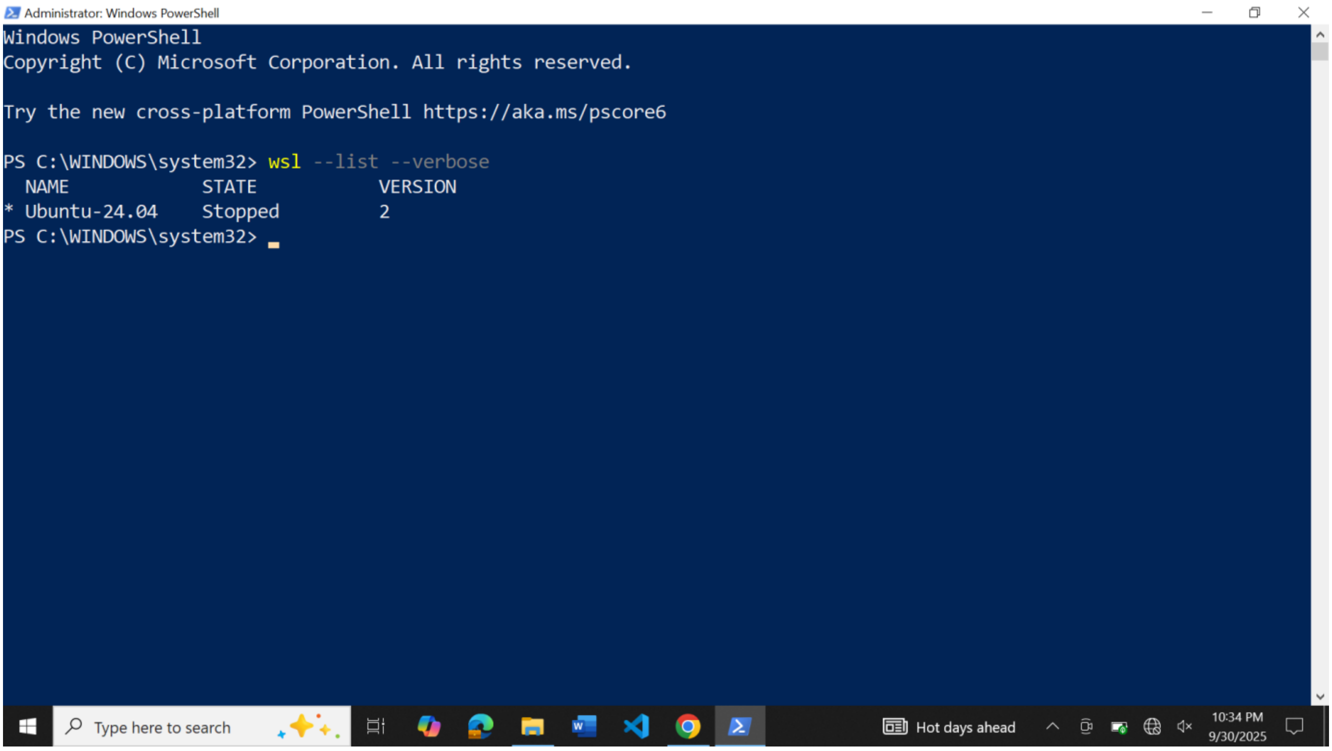
Part A: WSL2 & Ubuntu Setup

1. **Verify WSL2 and Ubuntu installation**
   - Verify installation by running the following command in powershell:

   ```
   wsl --list --verbose
   ```

   Submit a screenshot showing Ubuntu installed and running on WSL2.
2. **Update Ubuntu environment**
   - Run the following command in Ubuntu:

   ```
   sudo apt update && sudo apt upgrade -y
   ```



Part B: Git & GitHub SSH Setup

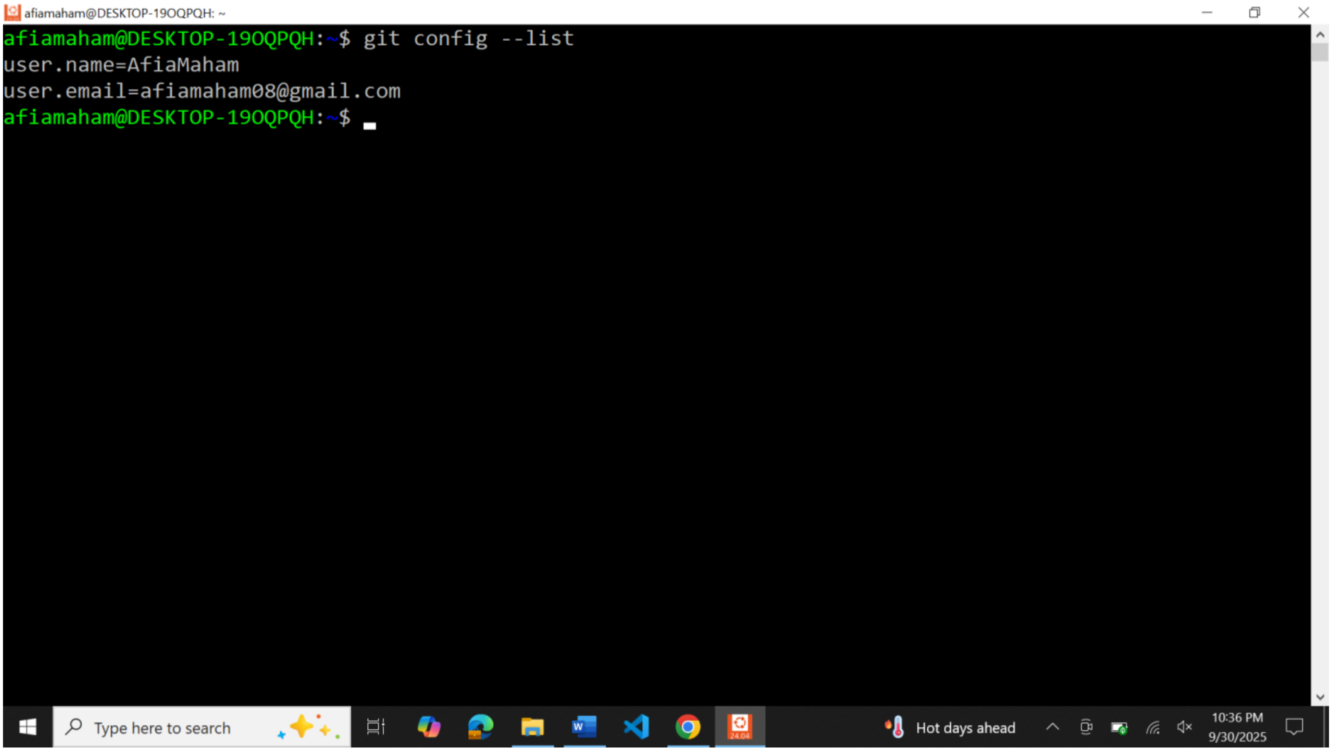
1. **Configure Git**
   - Set your name and email:

```
git config --global user.name "Your Name"
git config --global user.email "your@email.com"
```

   - Show your config:

```
git config --list
```

Submit a screenshot.



2. **Generate SSH Keys**
   - Run:

     ```
     ssh-keygen -t ed25519
     ```

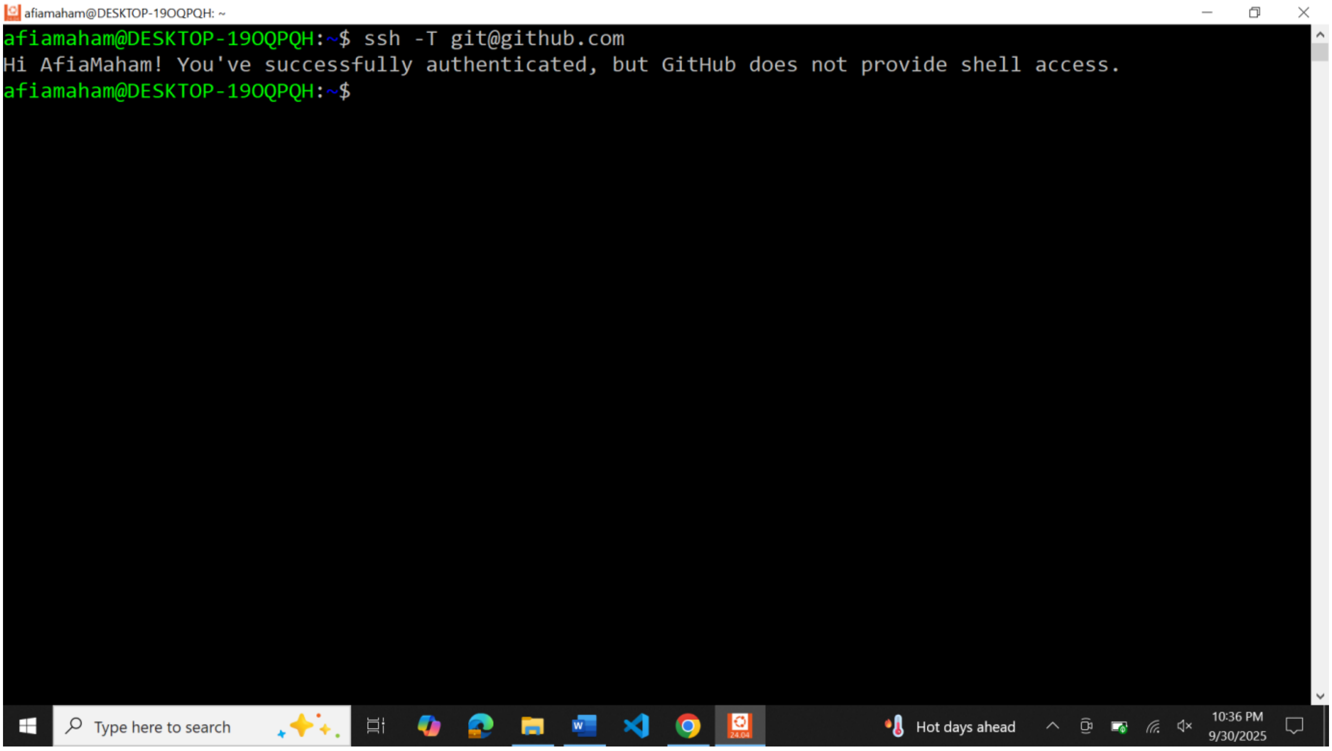
   - Copy the public key:

     ```
     cat ~/.ssh/id_ed25519.pub
     ```

   - Add this key to your GitHub account under **Settings** → **SSH and GPG keys**.
3. **Test Connection**

   ```
   ssh -T git@github.com
   ```

   - Submit a screenshot showing successful authentication.



Part C: C Programming Environment & Practice

Step 1: Install Build Tools

Before writing C programs, install the **build-essential** package which contains `gcc`, `g++`, and other tools required for compiling.

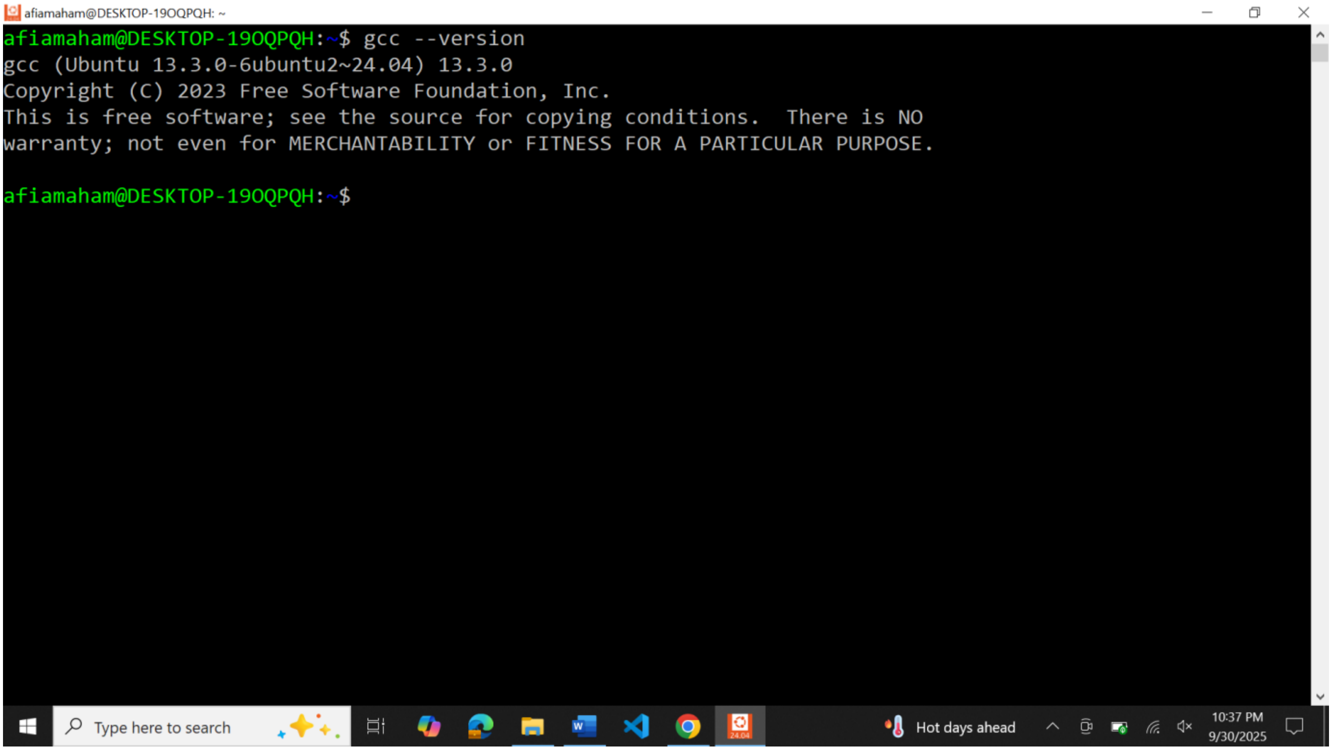
Run:

```
sudo apt install build-essential
```

Verify installation by checking the version of `gcc`:

```
gcc --version
```

- Submit a screenshot of successful installation and version output.



Step 2: How to run a C Program

1. First write a C program in a file with `.c` extension.
2. Compile the file using `gcc filename.c -o filename.out`
3. Execute it using `./filename.out`

Breakdown

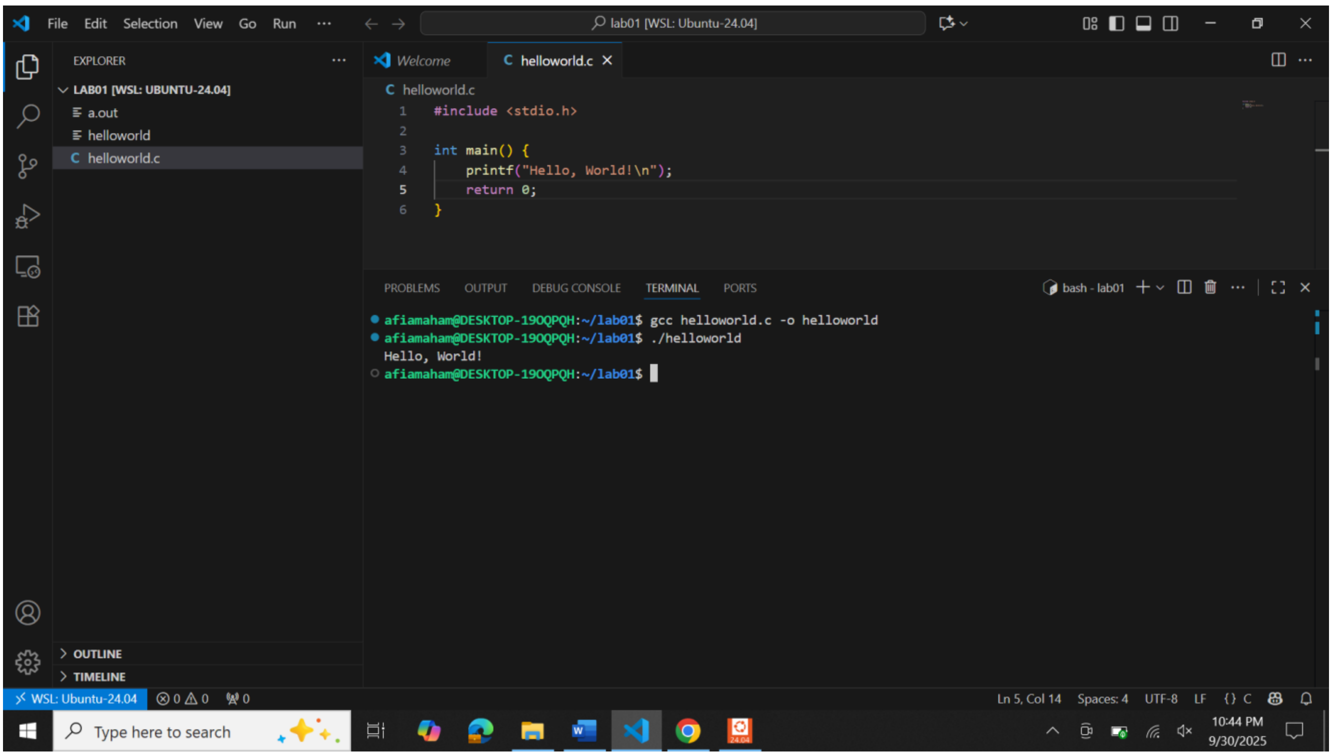
- `gcc`
  - This is the GNU Compiler Collection command.

    - It compiles C (and other languages like C++) programs into machine code that can be executed by the computer.
- `filename.c`
    - This is the source code file you wrote in C.
    - Example: hello.c contains your C program.
- `-o filename.out`
    - The option -o means “output.”
    - By default, gcc creates an executable file named a.out if you don’t specify anything.
    - With -o, you can choose the name of the output executable.
    - In this case, the compiled file will be named filename.out.

Step 3: Write a C Program

Write a simple C program of your choice. It can be a **Hello World** program or any other.

> **Submission:** For the program, submit:
>
> - The C source code (`.c` file).
> - Screenshot of execution (`./program`)



Deadline

- Submit before **12:00 AM, 25 September, 2025**.